Table 1: Baseline Characteristics of the Study Population, Overall and by Surgical Start Time

Variable	Overall N = 31,779[1]	6-8am N = 11,117[1]	8-11am N = 7,896[1]	11-1pm N = 5,579[1]	1-3pm N = 4,664[1]	3-5pm N = 2,005[1]	5-7pm N = 518[1]	p-value[2]
Age	58 ± 15	58 ± 14	58 ± 15	58 ± 15	58 ± 15	58 ± 15	56 ± 16	0.021
Risk Stratification Index (30-day mortality)	-0.53 ± 1.04	-0.64 ± 1.04	-0.59 ± 1.05	-0.48 ± 1.03	-0.41 ± 0.99	-0.25 ± 1.02	-0.10 ± 0.96	<0.001
Charlson Comorbidity Index	1.19 ± 1.88	1.15 ± 1.79	1.13 ± 1.88	1.23 ± 1.97	1.30 ± 1.94	1.26 ± 1.91	1.21 ± 1.99	<0.001
Gender								<0.001
Male	17,115 (54%)	5,559 (50%)	4,611 (58%)	3,088 (55%)	2,469 (53%)	1,105 (55%)	283 (55%)	
Female	14,664 (46%)	5,558 (50%)	3,285 (42%)	2,491 (45%)	2,195 (47%)	900 (45%)	235 (45%)	
ASA Physical Status								<0.001
I-II	17,079 (54%)	6,284 (57%)	4,250 (54%)	2,874 (52%)	2,369 (51%)	1,036 (52%)	266 (51%)	
III	13,647 (43%)	4,574 (41%)	3,381 (43%)	2,481 (44%)	2,120 (45%)	871 (43%)	220 (42%)	
IV-VI	1,053 (3.3%)	259 (2.3%)	265 (3.4%)	224 (4.0%)	175 (3.8%)	98 (4.9%)	32 (6.2%)	
Procedure								<0.001
Other non-major procedures	4,291 (14%)	1,144 (10%)	1,249 (16%)	863 (15%)	634 (14%)	318 (16%)	83 (16%)	
Gynecologic	4,512 (14%)	1,326 (12%)	1,217 (15%)	824 (15%)	735 (16%)	317 (16%)	93 (18%)	
Urologic	6,639 (21%)	2,872 (26%)	1,266 (16%)	891 (16%)	1,132 (24%)	400 (20%)	78 (15%)	
Orthopedic & Spine	10,848 (34%)	4,286 (39%)	2,718 (34%)	1,883 (34%)	1,326 (28%)	528 (26%)	107 (21%)	
Gastrointestinal & Hernia	5,489 (17%)	1,489 (13%)	1,446 (18%)	1,118 (20%)	837 (18%)	442 (22%)	157 (30%)	
Cancer	10,942 (34%)	3,900 (35%)	2,567 (33%)	1,798 (32%)	1,802 (39%)	727 (36%)	148 (29%)	<0.001
Cardiovascular/Cerebrovascular disease	16,162 (51%)	5,659 (51%)	3,980 (50%)	2,860 (51%)	2,390 (51%)	1,022 (51%)	251 (48%)	0.8
Dementia	241 (0.8%)	58 (0.5%)	63 (0.8%)	45 (0.8%)	55 (1.2%)	16 (0.8%)	4 (0.8%)	0.002
Diabetes	4,163 (13%)	1,474 (13%)	1,006 (13%)	776 (14%)	605 (13%)	247 (12%)	55 (11%)	0.14
Digestive disease	7,020 (22%)	2,422 (22%)	1,765 (22%)	1,252 (22%)	1,034 (22%)	441 (22%)	106 (20%)	0.8
Osteoarthritis	5,711 (18%)	2,167 (19%)	1,684 (21%)	1,006 (18%)	622 (13%)	199 (9.9%)	33 (6.4%)	<0.001
Psychiatric disorder	2,904 (9.1%)	962 (8.7%)	728 (9.2%)	577 (10%)	410 (8.8%)	186 (9.3%)	41 (7.9%)	0.012
Pulmonary disease	3,467 (11%)	1,150 (10%)	865 (11%)	633 (11%)	517 (11%)	241 (12%)	61 (12%)	0.2
30-day Mortality	138 (0.4%)	32 (0.3%)	25 (0.3%)	36 (0.6%)	21 (0.5%)	19 (0.9%)	5 (1.0%)	<0.001

[1] Mean ± SD; n (%)

[2] One-way analysis of means; Pearson's Chi-squared test; Fisher's Exact Test for Count Data with simulated p-value (based on 10000 replicates)

Criterion (AIC) was selected. Using this optimal spline structure, each variable was then tested for nonlinearity using Wald χ^2 statistics, and spline terms were retained only when the nonlinearity was statistically significant.

For the primary exposure (hour of surgery), both spline-based and categorical parameterizations were considered. Categorical modeling was prespecified using six clinically meaningful intervals (6–8 am, 8–11 am, 11 am–1 pm, 1–3 pm, 3–5 pm, and 5–7 pm), following recommendations to use ≤4-hour time blocks aligned with circadian patterns.[1] This approach also allowed comparison with prior studies, such as Sessler et al., which treated hour as a continuous exposure.

Associations with mortality were reported as odds ratios (ORs) with 95% confidence intervals (CIs). For the primary exposure, predicted 30-day mortality percentages from unadjusted and adjusted models were additionally examined.

Results

The final analytic cohort included 31,779 adult patients (mean±SD age, 58±15 years; 14,664 [46%] female). Patients younger than 18 years (n=212) were excluded because the original study population and associated risk indices were developed for adult surgery, along with 2 patients missing gender and 8 missing ASA physical status. Table 1 summarizes baseline characteristics by surgical start time. Surgery hour was categorized into six intervals (6–8 am, 8–11 am, 11–1 pm, 1–3 pm, 3–5 pm, and 5–7 pm). Most operations occurred during the morning hours (6–11 am), with substantially fewer surgeries performed later in the day(5-7 pm), reflecting typical scheduling and operating room workflows. Significant differences across groups were observed in 30-day mortality and several covariates, including age, RSI, Charlson Comorbidity Index, gender, ASA physical status, procedure type, and selected comorbidities (cancer, dementia, osteoarthritis, and psychiatric disorder).

A LOESS assessment suggested potential nonlinear associations for age, RSI, and hour of surgery. When these variables were modeled jointly using restricted cubic splines, a 3-knot specification provided the best fit (AIC=1242), compared with 4 knots (AIC=1249), 5 knots (AIC=1244), or a linear specification (AIC=1247). Subsequent tests showed strong evidence of nonlinearity for RSI ($p<0.001$), but not for age ($p=0.324$) or hour of surgery ($p=0.999$). Accordingly, RSI was modeled using a 3-knot spline, whereas age was modeled linearly and hour of surgery was modeled categorically.

In adjusted logistic regression analyses, RSI was strongly associated with 30-day mortality (overall $p<0.0001$). The spline demonstrated a sharply increasing risk beginning around RSI≈2, with predicted mortality exceeding 10% at the highest RSI levels (Figure 1).

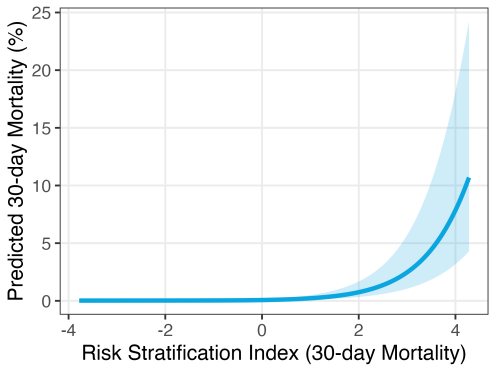

Figure 1: Adjusted Spline Relationship Between the Risk Stratification Index and Predicted 30-Day Postoperative Mortality

Hour of surgery was not associated with postoperative mortality in adjusted analyses (overall p=0.556). Compared with procedures performed between 8–11 am, surgeries conducted during later intervals (11 am–1 pm, 1–3 pm, 3–5 pm, and 5–7 pm) did not exhibit significantly different odds of death (Figure 3). In unadjusted analyses, predicted 30-day mortality increased throughout the day, from approximately 0.29% during 6–8 am to 0.97% during 5–7 pm (Figure 2). After adjustment for demographic, clinical, and procedural factors, these differences were markedly attenuated. Adjusted predicted mortality remained uniformly low across all hour categories, ranging from 0.005% to 0.043%, with no meaningful variation by time of day. The highest adjusted estimate occurred during 8–11 am (0.043%), while all other intervals ranged from 0.005% to 0.007%, indicating no clinically meaningful association between surgery hour and 30-day mortality.

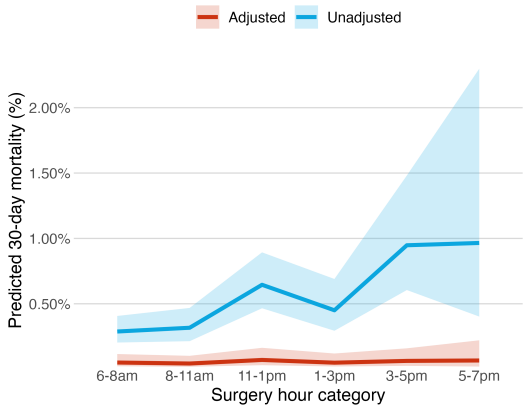

Figure 2: Comparison of Unadjusted and Adjusted Predicted 30-Day Postoperative Mortality by Surgical Start Time

ASA status was the strongest independent predictor of postoperative mortality. Compared with patients classified as ASA I–II, those with ASA IV–VI had substantially higher odds of death (OR, 9.84 [95% CI, 4.69–20.63]), as did those with ASA III (OR, 3.43 [95% CI, 1.77–6.65]). Osteoarthritis (OR, 2.24 [95% CI, 1.10–4.55]) and female sex (OR, 1.52 [95% CI, 1.03–2.24]) were also associated with increased mortality. Age was positively associated with risk (per-year OR, 1.02 [95% CI, 1.01–1.04]). No significant associations were observed for most other comorbidities or procedure categories.

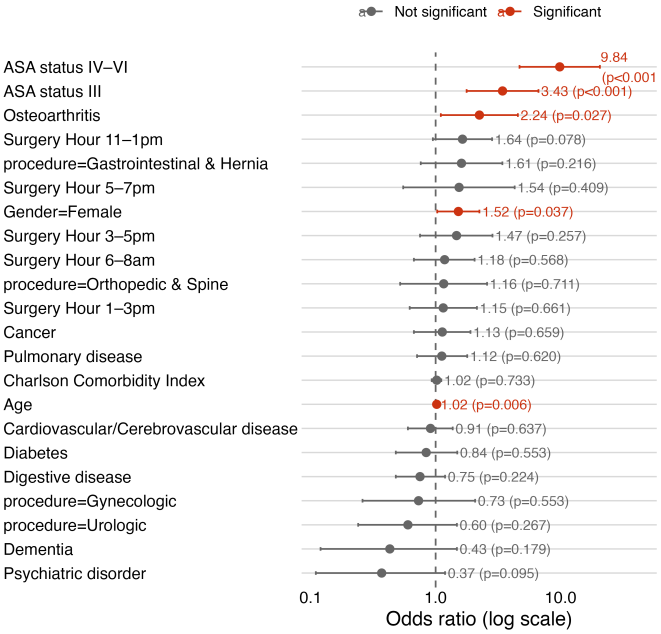

Figure 3: Adjusted Odds Ratios and 95% Confidence Intervals for Predictors of 30-Day Postoperative Mortality

Conclusion

After accounting for patient characteristics, comorbidities, and procedure type, the hour of surgery showed no meaningful association with 30-day postoperative mortality. The apparent increase in unadjusted mortality later in the day was explained by differences in patient risk profiles rather than timing itself. In contrast, ASA status and overall physiologic burden, as measured by the Risk Stratification Index, were the strongest predictors of mortality.

These findings suggest that, within this surgical environment, short-term mortality risk is driven primarily by patient health status rather than the specific time surgery begins. Limitations include the observational design, potential residual confounding, incomplete capture of operational factors such as staffing patterns or fatigue, and the use of data from a single health system, which may affect generalizability.

Association Between Time of Day of Elective General Surgery and 30-Day Mortality

Yanqi Huang, MS Student[1]

Abstract

This retrospective cohort study evaluated whether surgical start time is associated with 30-day postoperative mortality among 31,779 elective general surgery patients treated at the Cleveland Clinic from 2005 to 2010. Surgical start time was grouped into six intervals, and multivariable logistic regression adjusted for demographic, clinical, and procedural factors. Continuous covariates were assessed using LOESS and restricted cubic splines(RCS). Unadjusted mortality increased across the day (0.29% in 6–8 am vs. 0.97% in 5–7 pm), but after adjustment, start time was not associated with mortality (overall $p = 0.556$), and adjusted predicted mortality remained uniformly low (0.005%–0.043%). American Society of Anesthesiologists status(ASA) and risk stratification index(RSI) for 30-day mortality were the strongest predictors. These findings indicate that the apparent late-day increase in mortality is explained by differences in patient risk rather than surgical timing.

Keywords: surgical start time, postoperative mortality, logistic regression, restricted cubic splines

Introduction

Elective operations can be scheduled flexibly throughout the day because delaying urgent or emergency procedures can introduce substantial risks. Nonetheless, the time of day at which surgery is performed may influence postoperative outcomes. Circadian rhythm disruption during the perioperative period may reduce physiologic resilience,[1] and fatigue among surgeons and operating room staff may impair performance as the workday progresses.[2, 3, 4, 5]

However postoperative mortality is strongly shaped by patient-level risk factors. Physiologic status and comorbidity burden—captured by the ASA physical status classification, RSI(30 day mortality), and older age—are among the most consistent predictors of short-term mortality and may vary by time of day due to scheduling patterns. Evaluating whether surgical start time is associated with mortality therefore requires careful adjustment for these factors to distinguish true timing effects from underlying patient complexity.

[1]Department of Biostatistics and Informatics, Colorado School of Public Health, University of Colorado Anschutz Medical Campus, Aurora, CO

Research Question

Is surgical start time associated with 30-day postoperative mortality after accounting for major patient-level risk factors such as ASA status, comorbidity burden, RSI and age?

Materials and Methods

This retrospective cohort study used data from the Teaching Statistics in the Health Sciences (TSHS) repository, originally derived from Sessler et al. (2011), containing 32,001 elective general surgery patients treated at the Cleveland Clinic from 2005 to 2010.[2]

The primary outcome was 30-day postoperative mortality, and the primary exposure was the hour of surgery at case start. Covariates considered potential confounders included age, RSI, the Charlson Comorbidity Index, gender, ASA physical status, the Agency for Healthcare Research and Quality Clinical Classifications Software (AHRQ-CCS) procedure category, and baseline comorbidities (cancer, cardiovascular or cerebrovascular disease, dementia, diabetes, digestive disease, osteoarthritis, psychiatric disorder, and pulmonary disease).

Procedure types were grouped into clinically meaningful categories. Endocrine, breast, and plastic procedures, each contributing only one mortality event, were combined into an "other non-major procedures" category, resulting in five final groups: gastrointestinal and hernia, orthopedic and spine, gynecologic, urologic, and other non-major procedures.

Statistical Methods

All analyses were conducted in RStudio version 2025.09.2+418. Baseline characteristics were summarized using means and standard deviations for continuous variables and counts with percentages for categorical variables. Group differences were assessed using t tests or one-way ANOVA for continuous variables and chi-square or Fisher exact tests for categorical variables.

The association between surgical start time and 30-day mortality was evaluated using multivariable logistic regression including all prespecified covariates. The complication-specific RSI was excluded because of high correlation with the mortality-specific RSI. Clinically standard reference categories were used (e.g., 8–11 am for start time, ASA I–II for ASA status, and male sex for gender).

Continuous covariates were assessed for nonlinearity using LOESS and RCS.[6] RCS are recommended to minimize residual confounding, with 3–5 knots commonly used.[7] Candidate nonlinear variables were modeled jointly using RCS. Models with 3, 4, and 5 knots were compared against a fully linear specification, and the number of knots with the lowest Akaike Information

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